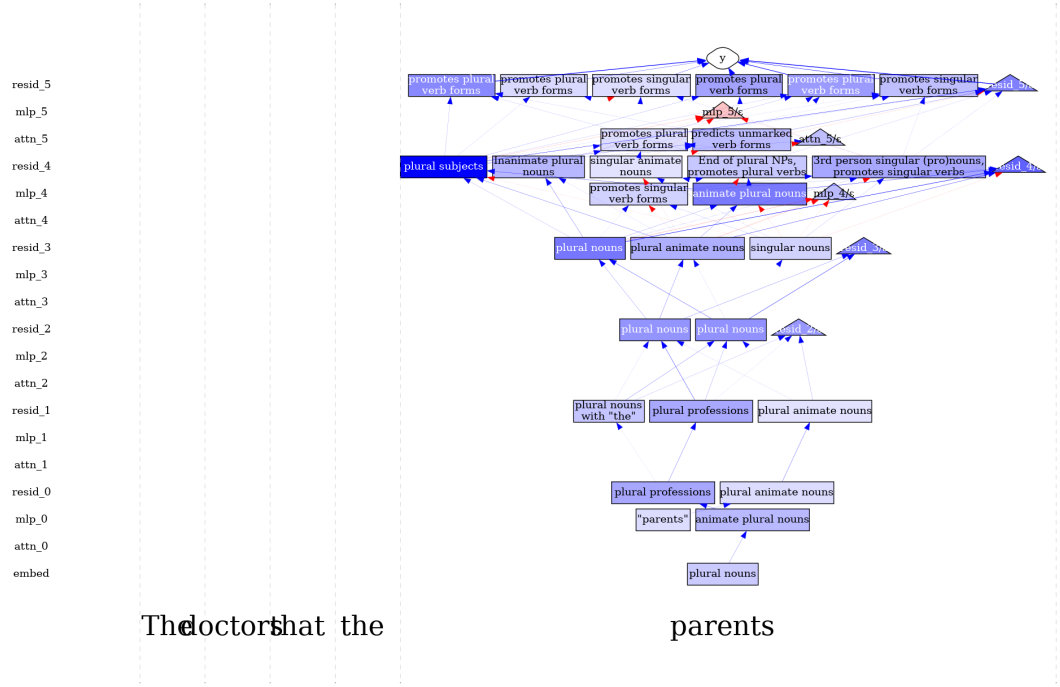

Figure 13: The feature circuit for agreement within a relative clause in Pythia-70M, computed with $T_N = 0.2$ and $T_E = 0.02$. The model detects the subject's number at the subject (within the RC)'s position in early layers. In later layers, these features are inputs to features which activate on anything predictive of particular verb inflections.

run the SHIFT with neurons baseline. We therefore instead compare to the neuron skyline, where we allow the skyline an unfair advantage by simply ablating neurons which have positive effects on gender-based probabilities given the balanced set.

E IMPLEMENTATION DETAILS FOR CLASSIFIER EXPERIMENTS

E.1 CLASSIFIER TRAINING

Here we describe how we train linear classification heads on Pythia-70M and Gemma-2-2B the Bias in Bios (BiB) task of §4.

Given a model M and choice ℓ of layer, we mean-pool over (non-padding) tokens all layer ℓ residual stream activations from M; we then train a linear classification head via logistic regression, using the AdamW optimizer (Loshchilov & Hutter, 2017) and learning rate 0.01 for one epoch on this dataset of activations. The activations and labels for this logistic regression are collected from the **ambiguous set** for the baseline classifier and from the **balanced set** for the oracle classifier.

To mimick a realistic application setting, we tune the choice ℓ of layer for the baseline probe's accuracy on (a test split of) the **ambiguous set**. For Pythia, this recommends using the penultimate layer $\ell = 4$. For Gemma, there is a wide range of equally performant layers. Thus—for this one choice only—we make use of the **balanced** set to compute how the baseline probe generalizes; we select the layer $\ell = 22$ for which the baseline probe generalizes *worst*. We make this choice to set up a testbed where there is the most space for improvement. We emphasize that we never tune hyperparameters for the performance **balanced set** of SHIFT, as using the **balanced set** is forbidden by the problem statement.

When retraining after performing SHIFT, we retrain *only* the linear classification head, not the full model.

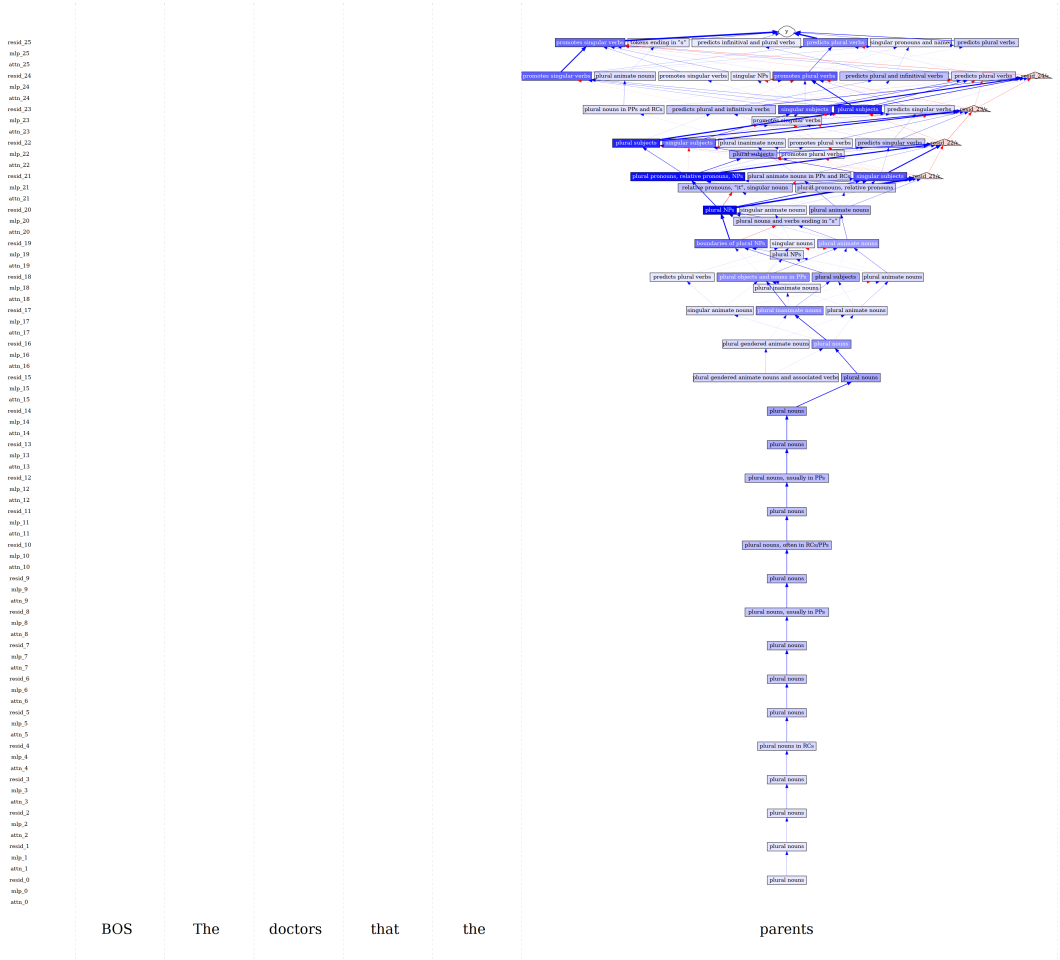

Figure 14: The feature circuit for agreement within a relative clause for Gemma-2-2B, computed with $T_N = 0.5$ and $T_E = 0.05$.

photographs 17.43	now covered with black bars. Mann said he used the image without permission “to illustrate that this is the kind of thing that shouldn’t be shown”. Mann claimed that after Virginia saw the article, she started touching herself on the areas that were blacked out, saying, “what’s wrong with me?” Mann responded to the criticisms saying she did not plan the photographs and that when she was young, she was often nude, so she raised her children similarly.↵↵Many of her other photographs containing her nude or hurt children caused controversy. For example, in The Perfect Tomato, the viewer sees a nude Jessie, posing on
motions 16.97	(1955).↵Nothing contained in the affidavits indicates the jury considered any extraneous facts. All the statements in the affidavits relate to evidence that was presented at trial or to information that was otherwise brought to the attention of the jury during the trial. The affidavits provide no evidence that the jury consulted any outside sources of information regarding the definition of "standard of care," or regarding any other matter. Nothing in either of the affidavits indicates that the jury, or any particular juror, was influenced by any outside source. The trial court did not abuse its discretion in denying the plaintiffs' posttrial motions seeking discovery regarding the jury's deliberations
dictions 16.55	the relevant↵jurisdictions (New York and
given 13.97	agreement with percolation thresholds given in Table \[
defs 13.71	to "template typedefs", you can declare

Figure 15: An example sparse feature for agreement across a relative clause in Gemma 2 (`resid_12/13561`). This feature activates on tokens in noun phrases where the noun head is plural, but not on singular distractor phrases within the plural NP. This feature carries the number of the subject across positions, so we term it an “NP number tracker”.

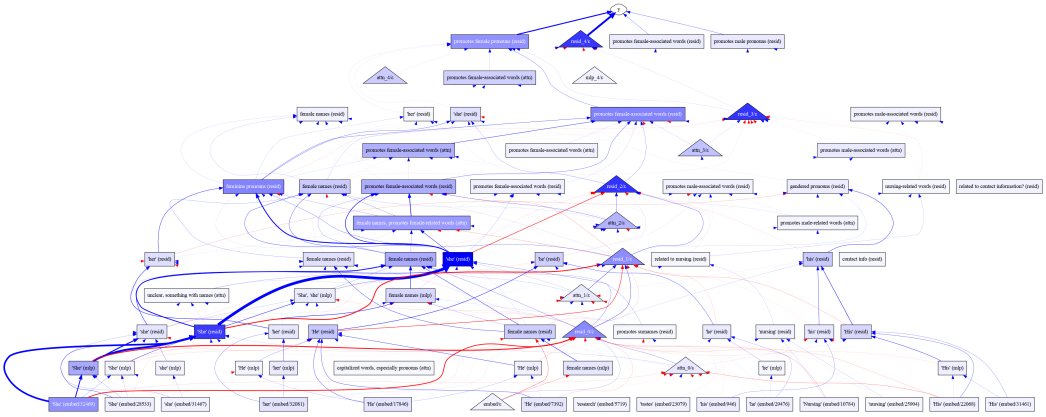

Figure 16: The full annotated feature circuit for the Bias in Bios classifier. Many nodes simply detect the presence of gendered pronouns or gendered names. A few features attend to profession information, including one which activates on words related to nursing, and another which activates on passages relating to science and academia.

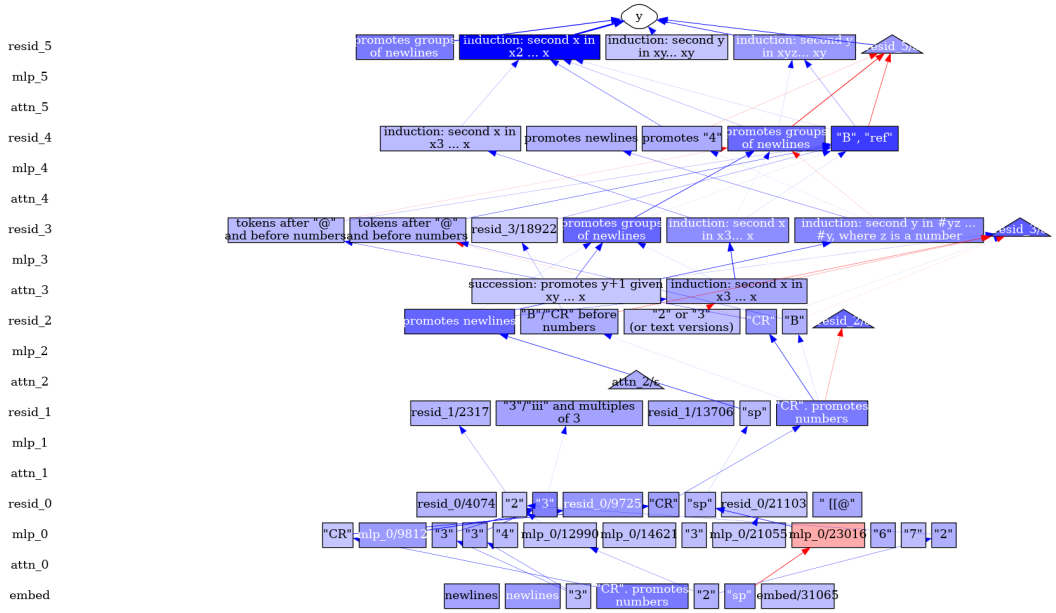

Figure 17: The full annotated feature circuit for incrementing number sequences. The model first detects the presence of specific number tokens, like “3”. Later, it learns more robust semantic representations of those numbers, like “iii” and “Three”. Then, the model uses a series of narrow and general succesion and induction features to increment the next number.